Shenzhen Hi-Link Electronic Co., Ltd.

HLK-LD6004

3D Presence Radar Module
Technical Specification

Version: V1.0 — September 2025

Shenzhen Hi-Link Electronic Co., Ltd.

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1. Product Overview

The HLK-LD6004 is a radar sensing module developed by Hi-Link based on the **ADT6101P** chip. It integrates a **57–64 GHz** RF transceiver system, a **2T2R AIP** packaged antenna, **1 MB Flash**, a radar signal processing unit, and an **ARM® Cortex®-M3** core.

This module operates on the **FMCW** (Frequency-Modulated Continuous Wave) signal processing mechanism. Combined with radar signal processing algorithms, it achieves high-sensitivity human body state detection — capable of identifying **movement, micro-motion, and stationary human bodies**, and calculating target distance, velocity, and other auxiliary information.

The module uses a two-transmit, two-receive antenna configuration with a wide beam angle, making it suitable for **ceiling installation**. It supports **GPIO** and **UART** signal output for flexible integration into various intelligent scenarios.

2. Product Features

- FMCW radar detection for high-sensitivity presence sensing
- Wide detection angle: horizontal ±60°, vertical ±60°
- Supports wall-mount mode switching with maximum sensing distance of 6 m
- Tracks up to 3 simultaneous targets with X, Y, Z coordinate output
- Detects both moving and stationary persons indoors; supports configurable detection zones and interference shielding
- Standard UART interface with published communication protocol
- Adjustable UART parameters to meet different scene requirements
- Compact size: 25 × 31.5 mm; supports pin-header and SMT connection methods
- Immune to temperature, humidity, noise, airflow, dust, and ambient light

3. Application Scenarios

Smart Home Appliances: Detects room occupancy in real time; adjusts appliance modes (working / standby / off) to achieve intelligent automation.

Human-Sensing Lighting: Senses human presence in a space; automatically controls lighting — public areas, offices, all types of presence-sensing lights.

Smart Home / Hotel / Office: Real-time detection of whether a person has entered or is present in a location (home, hotel, office, bathroom); enables security, appliance control, and personnel monitoring. Integrates with IoT platforms.

4. Electrical Characteristics and Parameters

4.1 Functional Parameters

Parameter	Min	Typical	Max	Unit	Notes
Max detection targets			3	persons	
Distance accuracy		0.4		m	
Wall-mount sensing distance	0		6	m	
Recognition accuracy		95		%	

4.2 Electrical Characteristics

Parameter	Min	Typical	Max	Unit
Operating Voltage (VCC)	3.1	3.3	3.5	V
Operating Current (ICC)		135	600	mA
Operating Temperature (TOP)	-20		85	°C
Storage Temperature (TST)	-40		85	°C

4.3 RF Characteristics

Parameter	Min	Typical	Max	Unit
Operating Frequency	58		64	GHz
Transmit Power (EIRP)		12		dBm
Antenna Gain		4		dBi
Horizontal Beam (–3 dB)	–60		+60	°
Vertical Beam (–3 dB)	–60		+60	°

5. Hardware Description

5.1 Physical Dimensions

Overall board size: **35.00 mm (L) × 7.00 mm (W)**. Pin pitch: **1.27 mm**. Antenna zone centered at 17.00 mm from left. The module origin (0,0,0) is located at the antenna center.

5.2 Pin Definition

Pin Name	Description	Notes
3.3V (×2)	Power input 3.3V	Both sides of the connector
OUT	GPIO P20 / UART2 TXD	HIGH when person detected in zone; LOW when no person
RX0	GPIO P01 / UART0 RXD	UART receive (host → module)
TX0	GPIO P00 / UART0 TXD	UART transmit (module → host)
GND (×2)	Ground	
SWCLK	SWD Debug Clock	For firmware debugging only
SWDIO	SWD Debug Signal	For firmware debugging only
BOOT1	Boot config pin	Must be pulled LOW before power-on

5.3 Module Peripheral Reference Design

Connect a 5-pin header (P2) as follows (pin 1 = bottom):

Pin	Signal
1	OUT (GPIO P20 — person detected indicator)
2	TX0 (UART0 TX)
3	RX0 (UART0 RX)
4	GND
5	3V3

5.4 Boot Configuration

	BOOT1	BOOT0	Mode
Level	0	1	Boot from internal Flash (normal operation)
Pin No.	Pin 8	Pin 12	

Note: BOOT1 and BOOT0 both have internal pull-ups. BOOT1 **must be pulled LOW before power-on** for normal Flash boot.

6. Usage and Configuration

6.1 Typical Application Circuit

The HLK-LD6004 can be used directly by reading the OUT/TX2 pin (HIGH = person detected, LOW = no person). Simultaneously, UART0 outputs structured detection results per the communication protocol, including target position, velocity, and auxiliary data.

• Supply voltage: **3.3V** with current capacity **>1A**.
• All IO outputs: **3.3V logic**.
• Default UART baud rate: **115200, 8N1** (no parity).

6.2 GUI Visualization Tool

Step 1 — Connect Device:

Set baud rate to 115200 → select the corresponding COM port → click [Start]. Module begins detection.

Step 2 — Set Detection Zones:

Set detection zone coordinates (module is the origin). Up to 4 detection zones can be configured. For rooms with ceiling air conditioners, limit the Z-axis range to filter out AC interference. Click [Set Detection Zone]. Settings persist across power cycles.

Step 3 — Set Interference Zones:

Set coordinates for areas that may cause false detections (e.g., AC units, fans, curtains). Targets within interference zones do not trigger OUT/TX2 output. Up to 4 interference zones. Click [Set Interference Zone]. Persists across power cycles.

Step 4 — Set Stay Zones:

For demanding scenarios (deep sleep detection, glass-partitioned bathrooms), configure stay zones with independent delay times separate from the main detection zones. Click [Set Stay Zone]. Persists across power cycles.

Other GUI Functions:

Function	Description
Firmware Version	Auto-detected on connection
Installation Mode	Switch between ceiling mount and wall mount
Hold Delay	Time the 'person present' state is maintained after detection (default 30s)
Stay Zone Lifecycle	Independent delay for stay zones. Set lifecycle = time × 20 (default 300 = 15s)
UART Output Interval	Reporting interval; default 50ms/frame. Setting=10 → 500ms/frame
Low-Power Mode	Enabled: switches to 1T1R when no person (avg 2.5mA); normal 2T2R mode (avg 135mA). Sleep time ≤50
Radar Off / P20 HIGH	Disables radar; OUT pin held HIGH continuously
Radar Off / P20 LOW	Disables radar; OUT pin held LOW continuously
Strong Reflection Mode	For environments with many metal surfaces
P20 Output Mode	Default: HIGH when person / LOW when no person. Options: inverted, continuous HIGH/LOW, 500ms puls
Point Cloud Display	Toggle raw point cloud data visualization on/off
Target Display	Toggle tracked target point visualization on/off
Detection Sensitivity	Low/Medium/High. Low = motion only; High = includes micro-motion
Trigger Speed	Delay from detection to output trigger: Fast=0.5s, Medium=1s, Slow=5s
Auto-Generate Interference Zone	With no persons present, activate interference sources; click to shield detected interference points

Function	Description
Clear Interference Zones	Removes manually set or auto-generated interference zones
Get Configured Zones	Retrieves currently saved detection and interference zone settings
Reset Detection Zones	Restores default 6×4m detection area
Reset No-Person State	Forces detection state to 'no person' immediately (useful for testing)
Z-Axis Range Setting	Sets overall Z detection range. Module antenna = origin (Z min = 0)

6.3 OTA Upgrade

OTA (Over-The-Air) firmware upgrades are supported. Refer to the separate *OTA Upgrade Tool User Manual V1.0* document for detailed instructions. OTA can be triggered via the Bluetooth app (HLKRadarTool) or via the desktop GUI.

6.4 Installation Modes and Detection Range

Wall/Side Mount Mode:

Install at a height of **1.5–2 m** on a wall, with the antenna face pointing into the room. Maximum sensing distance: **6 m**.

Ceiling Mount Mode:

Install on the ceiling facing downward. The module detects persons within the ±60° cone below. Best suited for rooms where the sensor is directly overhead.

Detection Range (Wall-Mount, typical):

State	Max Range
Moving person	~5–6 m
Stationary/seated person	~2–4 m (varies with environment)

Actual detection distance depends on target RCS (radar cross-section), environment, and obstructions. Variation within a range is normal.

7. Precautions

1. The radar module's detection range depends heavily on target RCS and environmental factors. Effective sensing distance may vary with environment and target; fluctuation within a range is normal.

2. The module has **strict power supply requirements**: input voltage 3.1–3.5V, ripple ≤50mV, current ≥1A. If using a DC-DC converter, switching frequency must be ≥2 MHz.

8. Radar Radome Design

A radar radome protects the antenna from rain, sunlight, wind, and other external elements. However, it affects radar performance in several ways: dielectric and reflection losses reduce effective radiated power; beam distortion alters the detection zone; shell reflections degrade TX/RX isolation; and phase changes from electromagnetic wave transmission affect angle measurement. Careful radome design is therefore essential.

Material Selection:

Choose materials with low dielectric constant (εr) and low dissipation factor (tan δ). Recommended common materials:

Material	Dielectric Constant (εr)	Dissipation Factor (tan δ)
Polycarbonate (PC)	2.9	0.012
ABS	2.0–3.5	0.0050–0.019
PEEK	3.2	0.0048
PTFE (Teflon®)	2.0	<0.0002
Acrylic (Plexiglass®)	2.6	0.009
Glass	5.75	0.003
Ceramic	9.8	0.0005

Material	Dielectric Constant (εr)	Dissipation Factor (tan δ)
PE	2.3	0.0003
PBT	2.9–4.0	0.002

Design Rules:

• Radome surface must be **smooth and uniform in thickness**.

• **Radome wall thickness** formula: T = N × c / (2f√εr), N = 1, 2, 3… where c = 3×10■ m/s, f = center frequency (60 GHz).

• **Air gap** between antenna and radome inner surface: d = N × c / (2f), N = 1, 2, 3… At 60 GHz: c/2f = 2.5 mm — the minimum recommended gap is multiples of 2.5 mm.

Revision History

Revision	Release Date	Description
V1.0	2025-09-25	Initial release

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